INCH-POUND

MIL-E-70694 (AR)
30 December 1988

MILITARY SPECIFICATION

ELECTRONIC CONTROL UNIT: 11784504

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Electronic Control Unit (ECU) part of the Improved Lifting Target Elevating Mechanism (ILTEM) and the Target Holding Mechanism/Tank Gunnery (THM/TG) for the Remote Target System (RETS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation. (See 6.2)

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Material, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	- Parts, Equipment and Tools for Army Material, Packaging of
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

11784504	- Electronic Control Unit
11784505	- Schematic Diagram, Electronic Control Unit
11784632	- Panel, Electronic

PACKAGING DATA SHEET

11784504	- Packaging of Electronic Control Unit

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Fabrication. The electronic control unit shall be manufactured in accordance with drawing 11784504 and drawings pertaining thereto.

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3.2 General specifications. Where applicable, the ECU shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances

b. Inorganic protective surface finishings

c. Workmanship

d. Part identification and marking

e. Soldering electrical connections

3.3 Environmental requirements.

3.3.1 Vibration.

3.3.1.1 Non-operating. The ECU shall meet the requirements of 3.5.1.1 through 3.5.2.7 after having been subjected to MIL-STD-810, method 514.3, procedure I-3.2.1.2 using figure 514.3-34 for one hour along the vertical axis.

3.3.2 Temperature.

3.3.2.1 Non-operating (storage). The ECU shall not suffer permanent damage when subjected to at least four hours at each temperature extreme of -40°F and +160°F and shall meet the requirements of 3.5.1.1 through 3.5.2.7 when returned to and thermally stabilized at ambient temperature (+60°F to +90°F).

3.3.2.2 Operating. The ECU shall meet the requirements of 3.5.1.1 through 3.5.2.7 when thermally stabilized at -25°F and 125°F.

3.3.3 Pressure and water tight test. The ECU shall not show any signs of bubbles from any surface part when submerged in a tank of water and dry nitrogen pressurized to 1.5 to 2 PSI for 5 minutes. It shall meet the requirements of 3.5.1.1 through 3.5.2.7.

3.4 Construction.

3.4.1 General description. The Electronic Control Unit (ECU) responds to target commands, operating relays that raise or lower an armored vehicle target and fire the gunfire simulator (GUFS). The ECU is a sealed operating unit containing circuit cards and is operated through connectors and switches on the front panel.

3.4.1.1 Front panel. The front panel contains seven connectors with rubber waterproof covers, two rotary switches, one toggle switch and a battery light indicator.

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3.4.1.1.1 Connectors. Four connectors are used to input target hit information from four hit sensors attached to the targets. The remaining three connectors are used for the visual hit indicator lamp, the smoke scorer unit, the gunfire simulator.

3.4.1.1.2 Rotary switches. The rotary switch marked "Function" has seven positions with the first position turning the ECU on and positions two through seven providing selectable modes varying target raising and lowering functions. The second rotary switch provides four positions to vary hit sensor sensitivity, depending on ammunition type used.

3.4.1.1.3 Toggle switch. The toggle switch provides a test capability of target raising and lowering and the visual hit indicator lamp operating functions.

3.4.1.1.4 Battery lamp indicator. The battery lamp indicator provides the battery voltage condition during target raising or lowering operations.

3.4.1.2 Electronic components. The inner electronics consists of three printed circuit cards and six relays.

3.4.1.2.1 Circuit cards. The three circuit cards consists of a logic control card, a hit amplification card, and a battery test card.

3.4.1.2.2 Relays. The six relays controls the hydraulic lifting unit, visual hit indicator lamp, smoke scorer, and the gunfire simulator.

3.4.2 Construction. The ECU shall conform to the requirements drawings and notes contained in technical data package 11784504.

3.5 Performance requirements.

3.5.1 Test switch.

3.5.1.1 Raise target. The test switch is placed in the raise position and a signal is generated to raise the target and to turn on the visual hit indicator lamp. The target will remain in a raised position until a lower target signal is received. The visual hit indicator lamp will remain lit for as long as test switch is held in position. The up-valve relay will disengage 2 $\pm$ 1 seconds after the motor relay disengages.

3.5.1.2 Lower target. The test switch is placed in the lower position and a signal is generated to lower the target. The down-valve relay shall disengage 2 $\pm$ 1 seconds after motor relay disengages.

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3.5.2 Function switches.

3.5.2.1 Position zero. Position zero will perform as an off switch and shall remove all electrical power from the ECU.

3.5.2.2 Position one. The target will be raised and lowered by remote signal and hit signals from the hit sensor will have no effect on the target position. The smoke scorer relay will activate for 2 ± 1 seconds after a hit.

3.5.2.3 Position two. The target will be raised and lowered by remote signal and hit signals from the hit sensor will lower the target until another signal to raise the target is received. The smoke scorer relay will activate for 2 ± 1 seconds after a hit.

3.5.2.4 Position three. The target will be raised and lowered by remote signal and hit signals from the hit sensor shall lower the target. The ECU will generate another signal 11 ± 2 seconds after receipt of the hit signal to raise the target. The smoke scorer relay will activate for 2 ± 1 seconds after a hit.

3.5.2.5 Position four. The target will be raised and lowered by remote signal and the visual hit indicator and smoke scorer relays will operate for 2 ± 1 seconds after a hit. Hit signals from the hit sensor will have no effect on target position.

3.5.2.6 Position five. The target will be raised and lowered by remote signal. Hit signals from the hit sensor will operate the visual hit indicator lamp and smoke scorer relays for 2 ± 1 seconds and lower the target until another signal to raise the target is received.

3.5.2.7 Position six. The target will be raised and lowered by remote signal. Hit signals from the hit sensor will operate the visual hit indicator lamp and smoke scorer relays for 2 ± 1 seconds. The ECU will generate another signal 11 ± 2 seconds after receipt of a hit signal to raise the target.

3.5.3 Lifting mechanism failure. Failure of the target to complete raising or lowering operation will result in the ECU generating a signal to shut off the lifting mechanism motor 11 ± 2 seconds after raising/lowering motion was initiated.

3.5.4 Battery check indicator. The voltage indicator light (LED) will indicate sufficient power is available during raising and lowering of the target to operate the lifting mechanism. The light will remain constant when voltage exceeds 10.5 volts and will flash when voltage is equal to 10.5 -0.1 volts.

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3.5.5 Target sensitivity limit switch. The target sensitivity limit switch will operate independently of the function switch or sensitivity switch setting and will enable the ECU to process hit signals when the target is in a raised position. Should the switch be shorted to ground the ECU shall not respond to hit signals.

3.5.6 Hit signal. An acceptable hit signal based upon the sensitivity switch setting from any one or combination of hit sensors will lower the target depending upon the function switch setting. The ECU shall meet the requirements in Table I.

3.5.7 Remote control operation. The ECU shall be capable of receiving a 12 volt -1.4 volt remote pulse (equal to or greater than 25 ms) command to raise or lower the target, and energize the hostile fire simulator.

3.5.8 Bad hit sensors. The ECU shall be capable of detecting a shorted or open sensor condition by lowering the target. The ECU shall not respond to any hit signals, or remote target commands, or test switch actions to raise the target.

3.5.9 25 ms output. When a hit is processed a 25 ms $\pm$ 5 ms pulse shall be applied to pin ST1-G.

3.6 First article. Unless otherwise specified in the contract or purchase order, a sample should be subjected to first article inspection (see 4.3).

3.7 Workmanship. The requirements of MIL-STD-454 and other applicable specification shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the

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contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Inspection provisions.

4.2.1 Submission of product. Unless otherwise specified by the contracting officer, inspection of lots shall be in accordance with the provisions of MIL-STD-105.

4.2.2 Examination and tests.

4.2.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the Quality Assurance Provision (QAP) listed in the technical data package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.2.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 4 herein. The tabulated classification of defects in 4.3.2.3 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.2.2.3 Functional tests. The requirements and tests in Table I shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (60°F to 90°F).

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TABLE I. Classification of defects
(100 percent inspection)

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURES
CRITICAL:	None		
MAJOR			
101	Test switch - raise Target	3.5.1.1	4.6.4.1.2
102	Test switch - lower target	3.5.1.2	4.6.4.2
103	Function switch position 1	3.5.2.2	4.6.4.11
104	Function switch position 2	3.5.2.3	4.6.4.12
105	Function switch position 3	3.5.2.4	4.6.4.13
106	Function switch position 4	3.5.2.5	4.6.4.14
107	Function switch position 5	3.5.2.6	4.6.4.14
108	Function switch position 6	3.5.2.7	4.6.4.14

4.2.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample to comply with any of the stated requirements.

4.2.3.1 Acceptance and rejection. Rejected lots shall be screened for all defective characterisitcs. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 First article (initial production) submission. The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) ECUs manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production. The three (3) ECUs shall be tested as specified herein for all the requirements of this specification.

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4.3.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.3.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.4 Special sampling. One ECU shall be selected at random by a Government representative as a control sample from each 25 produced. The control sample shall meet the requirements and tests in Table II. Three ECUs shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table III.

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TABLE II. Special sampling

CONTROL TEST		REQUIREMENT	TEST PROCEDURES
301	Fabrication	3.1	Applicable drawings - Visual
302	General specifications	3.2	MIL-F-13926 Visual
303	Lifting mechanism failure	3.5.3	4.6.4.3
304	Battery check indicator	3.5.4	4.6.4.4
305	Remote control operation	3.5.7	4.6.4.5 through 4.6.4.7
306	Bad hit sensors	3.5.8	4.6.4.8 through 4.6.4.10
307	25ms output pulse	3.5.9	4.6.4.11.3
308	Bad target sensitivity limit switch	3.5.5	4.6.4.12.1
309	Amplifier sensitivity	3.5.6	4.6.4.15

TABLE III. Special sampling, environmental

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURES
310	Vibration (nonoperating)	3.3.1.1	4.6.2.1
311	Low temperature (operating)	3.3.2.2	4.6.3.1
312	High temperature (operating)	3.3.2.2	4.6.3.2
313	Pressure & water tight test	3.3.3	4.6.3.3

4.4.1 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607. Prior to acceptance inspection, equipment designs shall be submitted to and approved by the Government. Requests for inspection equipment approval see 6.3.

4.5.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

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4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or QAP forming a part of the contract shall be supplied by the contractor in accordance with applicable data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment. Proper operation of test equipment, test fixtures shall be verified before using same to run diagnostic tests.

4.6 Methods of inspection.

4.6.1 Environmental. Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with Test Facilities and Apparatus requirements of MIL-STD-810. To facilitate ease in testing the ECU, a simple test fixture should be built to simulate relay responses using Light Emitting Diodes (LEDs), lights, etc. and that bad sensors (open or shorted) and target reed limit switches be simulated using toggle switches. Remote control command signals could be simulated using push button switches wired to produce the appropriate responses.

4.6.2 Non-operating.

4.6.2.1 Vibration. This test shall be performed as specified in MIL-STD-810, Method 514.3, Procedure I-3.2.12 using Fig 514 -34 and one hour vertical axis. After vibration the assembly shall be tested as specified in Table I to determine compliance with 3.3.1.1.

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4.6.2.3 Temperature (high). Temperature shall be as specified in MIL-STD-810, Method 501.2, Procedure I, Table 501 -I (2 cycles). After temperature testing, the assembly shall be tested as specified in Table I to determine compliance with 3.3.2.1.

4.6.2.4 Low temperature. Test shall be as specified in MIL-STD-810, Method 502.2, Procedure I-Storage, using 24 hours after thermal stabilization at -65°F. After temperature testing, the assembly shall be tested as specified in Table I to determine compliance with 3.3.2.1.

4.6.3 Operating.

4.6.3.1 Low temperature. Test shall be as specified in MIL-STD-810, Method 502.2, Procedure II, stabilize at -25°F for 24 hours. While at -25°F, the assembly shall be tested as specified in Table I to determine compliance with 3.3.2.2.

4.6.3.2 High temperature. Test shall be performed as specified in MIL-STD-810, Method 505.2, Procedure I. The assembly shall be tested as specified in Table I to determine compliance with 3.3.2.2 during the last cycle of the seven cycles.

4.6.3.3 Pressure and watertight test. This test shall be performed as follows. The ECU to be sealed and all parts plugged except the plug with performed packing, in which a nipple is to be installed for connection of air supply. Apply dry nitrogen air pressure for 5 minutes at 1.5 to 2 psi. (Do not exceed an upper limit of 3 psi as damage may occur). Submerse the ECU into a tank of water for 5 minutes and look for signs of bubbles escaping from the connectors, plugs, switches, screws, cover and welds. Any sign of air bubbles warrents a rejection. After pressure and watertight testing the ECU shall be subjected to the test of 4.6.3 for compliance with 3.3.3.

4.6.4 Functional. Tests contained in 4.6.4.1.1 through 4.6.4.15.3 are to be conducted utilizing appropriate calibrated meters, fixtures, adapters, and cables connected to the contacts specified in the TDP or respective test requirement paragraph. To facilitate ease in testing the ECU, a simple test fixture should be built to simulate relay responses using LEDs, lights, etc. and that bad sensors (open or shorted) and target reed limit switches be simulated using toggle switches. Remote control command signals could be simulated using push button switches wired to produce the appropriate responses.

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4.6.4.1 Target test switch - raising target and lamp test.

4.6.4.1.1 Initial conditions and switch conditions. 5.6 kilohms, 1/2w, 20% resistors across all amplifier inputs. (Simulates typical crystal resistance.) Function selector switch-position 1. Sensitivity switch in any position. Target "down" limit switch is open (target-down position). Target "up" limit switch is shorted to ground (target down position). Target sensitivity limit switch in any position. Supply voltage +12v ± .1 volt.

4.6.4.1.2 Raise target. Momentarily place test switch on the front panel to raise target. The motor and up valve relays shall set and while the switch is in the "raise target" position, the lamp relay shall be engaged while test switch is held. Within 5 seconds ± 1 second after initiation, change first, the target "down" limit switch to its "shorted" position (target up position) and then the target "up" limit switch to its open position (target up position). When the "up" limit switch is changed the motor relay will immediately disengage and the "up valve" relay will disengage 2 ± 1. seconds after the motor relay disengages. It will meet the requirements of 3.5.1.1.

4.6.4.2 Target test switch - lowering target. With the target limit switches in position for a raised target momentarily place the test switch in the position to lower the target. The motor and down valve relays shall set. Within 5 seconds ± 1 second after initiation change first, the target "up" switch to it's "down target" position and then the "down" limit switch to it's "down target position". When the "down" limit switch is changed the motor relay will immediately disengage and the "down valve" relay will disengage 2 ± 1 second after the motor relay disengages. It shall meet the requirements of 3.5.1.2.

4.6.4.3 Lifting mechanism failure. Repeat 4.6.4.1 and 4.6.4.2 (raising and lowering target using the test switch) but in each case do not change the respective limit switch positions. The motor relay shall disengage 11 ± 2 seconds after which the respective valve solenoid relay will disengage as well. It shall meet the requirements of 3.5.3.

4.6.4.4 Battery check indicator. Raise or lower the target using the front panel test switch. The front panel LED light shall be on continuously for as long as the motor relay is engaged and the ECU supply voltage greater than 10.5 volts. Raise or lower the target using the front panel test switch. During the next 5 seconds lower the supply voltage from the nominal + 12v to 8v and notice when the front panel led battery light starts to flash. The supply voltage battery test light shall start to flash when equal to 10.5 -0.1 volts. It shall meet the requirements of 3.5.4.

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4.6.4.5 Remote control operation - remotely raising target. With the switch positions set as indicated in 4.6.4.1.1 remotely simulate raising the target by momentarily pressing the remote raise push button switch. The relay actions and responses shall be as those in 4.6.4.1.2 except that the lamp relay shall not energize. It shall meet the requirements of 3.5.7.

4.6.4.6 Remote control operation - remotely lowering target. Remotely simulate lowering the target by momentarily pressing the remote lower push button switch. The relay actions and responses shall be as described in 4.6.4.2. It shall meet the requirements of 3.5.7.

4.6.4.7 Remote operation of hostile fire relay. While the hostile fire remote push button control switch is pressed, the hostile fire relay shall be energized. It shall meet the requirements of 3.5.7.

4.6.4.8 Bad hit sensor test - shorted crystal condition. Raise target using front panel test switch as described in 4.6.4.1. Using appropiately connected switches on amplifier inputs create a shorted crystal condition on connection to amplifier 1. Immediately when this happens the target should lower through activation of the motor relay and down valve relay. Restore a good crystal condition, raise the target and repeat this test for amplifiers 2 through 4. It shall meet the requirements of 3.5.8.

4.6.4.9 Bad hit sensor test - open crystal condition. Repeat the test in 4.6.4.8 but simulate an open crystal condition. It shall meet the requirements of 3.5.8.

4.6.4.10 Trying to raise target with a simulated bad crystal. With either a shorted or open crystal condition simulated, try to raise the target by remote and by the front panel test switch. The target shall not be raised by further means. It shall meet the requirements of 3.5.8.

4.6.4.11 Function switch test - position 1. Target remains up and does not lower.

4.6.4.11.1 Initial conditions switch and conditions. Function selector switch in position 1. Target sensitivity limit switch in the "open" position (ready to receive firing pulses-target up). Target "down" limit switch is open (target down position). Target "up" limit switch is shorted to ground (target down position). Sensitivity switch in position 1. Supply voltage + 12v $\pm$ .1 volt. Connect the pulse generator to amplifier #1 input and place a 5.6 kilohm 1/2w 20% resistor across all amplifier inputs to simulate

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good crystals. (Hit sensors) Set pulse generator to produce a 5 KHZ sine wave, having 100 mv peak amplitude set in a burst mode to produce a 2ms burst. Remove fault signal jumper for amplifier 1 (ST5-C to D).

4.6.4.11.2 Raise target. Raise the target remotely and set target limit switches accordingly. With the target raised burst the pulse generator. The smoke scorer relay shall activate for 2 ± 1 seconds but the target shall remain in the up position. It shall meet the requirements of 3.5.2.2.

4.6.4.11.3 25ms pulse output. Repeat the test using an oscilloscope connected to ST1-G. A 25ms ± 5ms pulse shall occur at the pin. It shall meet the requirements of 3.5.9.

4.6.4.12 Function switch test - position 2 - target lowers after a hit and stays lowered. Set the initial conditions per 4.6.4.11.1. With the target raised burst the pulse generator. The smoke scorer relay shall activate for 2 seconds ± 1 second and the target shall lower and stay lowered through activation of the motor and down valve relays. It shall meet the requirements of 3.5.2.3.

4.6.4.12.1 Bad target sensitivity limit switch. Repeat the test in 4.6.4.12 but short the target sensitivity limit switch in this case to ground. The target shall not lower nor shall the smoke scorer engage. It shall meet the requirements of 3.5.5.

4.6.4.13 Function switch test position 3 - target lowers after a hit and then automatically raises back up. Set the initial conditions per 4.6.4.11.1 and raise the target. Burst the pulse generator. The smoke scorer relay shall activate for 2 seconds ± 1 second and the target shall lower as indicated by the motor relay and the down-valve solenoid relays engaging. Switch target limit switches within 5 seconds accordingly for a lowered target by first changing the "up limit switch and then the "down" limit switch. The target shall raise again (motor and up valve relays engaging) 11 ± 2 seconds after the hit (pulse generator burst). Set target limit switches accordingly for a raised target by reversing the activation procedure ("down" switch first, then "up"). It shall meet the requirements of 3.5.2.4.

4.6.4.14 Function switch test positions 4, 5, and 6. Repeat the test procedure in 4.6.4.11 through 4.6.4.13 for positions 4, 5 and 6. The responses in positions 4, 5 and 6 shall be as those in positions 1, 2 and 3 except that in addition, the lamp relay shall activate for 2 ± 1 seconds at the same time as the smoke scorer relay. It shall meet the requirements of 3.5.2.5, 3.5.2.6 and 3.5.2.7, respectively.

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4.6.4.15 Amplifier sensitivity tests.

4.6.4.15.1 Initial conditions and panel switch conditions. Refer to 4.6.4.11.1 for initial conditions and front panel switch conditions with the exception that the function selector switch shall be in position 6.

4.6.4.15.2 Raise the target. Raise target using front panel test switch. Referring to Table IV, set the pulse generator to the frequency and front panel sensitivity switch positions accordingly. At each sensitivity switch position and frequency determine the lowest amplitude to a millivolt (MV) at which a "hit" will just be accepted and processed by the ECU amplifier. The millivolt reading shall lie within the range as specified in Table IV. It shall meet the requirements of 3.5.6.

4.6.4.15.3 Amplifiers 2-4. Repeat 4.6.4.15 for amplifiers 2-4. It shall meet the requirements of 3.5.6.

TABLE IV. ECU Sensitivity Response

2 MS Burst Frequency	Front Panel Sensitivity Switch Position	Peak Millivolt Acceptable Range
500 Hz	1	30 to 75 MV
500 Hz	2	95 to 140 MV
500 Hz	3	175 to 225 MV
500 Hz	4	205 to 255 MV
1000 Hz	1	10 to 55 MV
1000 Hz	2	50 to 95 MV
1000 Hz	3	95 to 140 MV
1000 Hz	4	115 to 160 MV
2000 Hz	1	5 to 45 MV
2000 Hz	2	25 to 70 MV
2000 Hz	3	55 to 100 MV
2000 Hz	4	70 to 110 MV
5000 Hz	1	5 to 50 MV
5000 Hz	2	25 to 70 MV
5000 Hz	3	50 to 95 MV
5000 Hz	4	60 to 105MV
10,000 Hz	1	10 to 55 MV
10,000 Hz	2	35 to 80 MV
10,000 Hz	3	60 to 105 MV
10,000 Hz	4	75 to 120 MV

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5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet 11784504 and MIL-P-14232. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. This specification details the specific requirements for performance, quality assurance, and testing for the Electronic Control Unit (ECU). The ECU will be used as part of the Tank Holding Mechanism/Tank Gunnery (THM/TG) and the Improved Lifting Target Elevating Mechanism (ILTEM) and will support live fire armor weapons training and qualification programs. These systems will provide training in identifying and firing on simulated armored vehicles during training exercises.

6.2 Acquistion requirements. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this and the item detail specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Block of serial numbers to identify each Distribution Box.

g. Requirement for first article submission.

6.2.2 Contract data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

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Reference Paragraph	DID Title	DID No.
4.6	Quality Engineering Acceptance Inspection Equipment Descriptive Documentation	DI-R-1714

(Data item descriptions related to this specification and identified in Section 6, will be approved and listed as such in DOD 5000.19L, Volume II, AMSDL. Copies of data item descriptions required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Submission of inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Subject term (key word) listing.

Electronic Control Unit (ECU)
Improved Lifting Target Elevating Mechanism (ILTEM)
Target Holding Mechanism/Tank Gunnery (THM/TG)
Remote Elevating Target System (RETS)

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 6920-A111)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-70694 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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